

Run one stored-program machine trace

Name _____

Opening prediction: The game score changes from 7 to 8. What inside the computer must have changed?

Word classification: mark each opening-board word **I** instruction, **D** data, or **N** not yet. memory[40] ____ memory[41] ____ memory[201] ____
Evidence _____

memory[80] = 0001 0100 11001000 (= 5320) **PC** = 80 **memory[200]** = 0000 0000 00000110 (= 6)

F fetch IR bits _____ IR whole-word decimal (= _____) PC _____ memory[200] bits _____ (= _____)	D decode operation _____ immediate _____ destination _____ control directs _____ memory[200] bits _____ (= _____)	E execute ALU work _____ + _____ = _____ control directs _____ PC _____ memory[200] bits _____ (= _____)	W write-back control directs result to [_____] PC _____ memory[200] bits _____ (= _____)
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Model boundary: One claim the four frames **support**: _____ One claim they **cannot** support: _____

Design budget: A universally higher clock is not the only use of the budget; designers may add _____ when the work can _____.

Change one condition: memory[200] starts as 0000 0000 00001001 (= 9).
Route/roles that stay the same: _____ New W memory bits _____ (= _____)

Boundary check: Write F, D, E, or W for each shuffled projected snapshot.

#1 _____	#2 _____	#3 _____	#4 _____
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One claim these four snapshots **cannot** establish about a real processor:

